

# The propositions package\*

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## Abstract

The `propositions` package provides a key-value driven system for labelling propositions, theses, and premises in academic papers. Items may be given names like ‘(P)’ or ‘Physicalism’, or auto-numbered using different counters; all carry robust cross-references with configurable formatting. The package integrates with `amsmath`, `hyperref`, and `cleveref`.

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## 1 Introduction

In some academic disciplines (such as philosophy), it is common to have displayed propositions (examples, theses, premises, . . . ) with various kinds of labels. A thesis might be referred to as ‘(P)’ or ‘Physicalism’; the premises of an argument might be numbered as ‘P1’, ‘P2’, ‘P3’, . . . ; or examples might be numbered consecutively over the course of a whole article. Standard L<sup>A</sup>T<sub>E</sub>X environments like `enumerate` can handle some of these cases, but cross-referencing is awkward: `\ref` produces a bare number or letter, and the author must manually add parentheses or other formatting at every point of reference. The standard `description` environment does not allow cross-referencing at all.

The `propositions` package solves this by attaching formatting information to each label. A short item like `\item[P]` (inside `prop`) is displayed as “(P)” and `\ref` automatically produces “(P)” as well—complete with parentheses, hyperlinks, and `cleveref` support. The full key-value interface supports named items, numbered items, custom counters, glosses, shorthands, and per-item format overrides.

The `\ptag` command (which requires `amsmath`) extends this to displayed math environments: an equation can be tagged with a proposition label instead of (or using) its equation number.

## 2 History

I wrote the ancestor to this package in the 90s while finishing my Ph.D. thesis, but never documented it or shared it with the world. This new version is a thorough re-implementation in L<sup>A</sup>T<sub>E</sub>X3, written in 2026 with extensive help from Claude Code. I hope others will find it as useful as I have.

## 3 Basic usage

Load the package with `\usepackage{propositions}` or `\usepackage[options]{propositions}` (see [section 9](#) below for valid package options).

The `prop` environment generates a list of propositions, each introduced by `\item`. `\item` with an optional argument gives a `description`-like label:

|                                                                                                                                                        |                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <pre>\begin{prop}   \item[Physicalism] Everything is   physical. \label{phys}   \item[Idealism] Everything is   mental. \label{ideal} \end{prop}</pre> | <pre><b>Physicalism</b> Everything is physi- cal. <b>Idealism</b> Everything is mental.</pre> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|

Unlike the standard `description` environment, one can refer back to these propositions using the standard `\ref` command:

|                                                           |                                                                       |
|-----------------------------------------------------------|-----------------------------------------------------------------------|
| <pre>\ref{phys} is more plausible than \ref{ideal}.</pre> | <pre><b>Physicalism</b> is more plausible than <b>Idealism</b>.</pre> |
|-----------------------------------------------------------|-----------------------------------------------------------------------|

With no optional argument, `\item` will by default generate numbered items similar to `enumerate`, but with numbering that persists across the document:

|                                                                                                                                                                                                                          |                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <pre> \begin{prop}   \item Every atom is physical.   \label{atoms} \end{prop} \ref{phys} follows from the conjunction of \ref{atoms} and \begin{prop}   \item Everything is an atom.   \label{atomism} \end{prop} </pre> | <p>(1) Every atom is physical.</p> <p><b>Physicalism</b> follows from the conjunction of (1) and</p> <p>(2) Everything is an atom.</p> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|

As with `enumerate`, the counter and formatting depend on the nesting level:

|                                                                                                                                                                                                                                                                                       |                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> \begin{prop}   \item \label{dual}   \begin{prop}     \item Some things are physical.     \label{some}     \item Some things are not     physical. \label{notall}   \end{prop} \end{prop} \end{prop} Without \ref{some}, \ref{dual} would be consistent with \ref{ideal}. </pre> | <p>(3) a. Some things are physical.</p> <p style="padding-left: 2em;">b. Some things are not physical.</p> <p>Without (3a), (3) would be consistent with <b>Idealism</b>.</p> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 4 Advanced usage

The format of the proposition labels, and of subsequent references, are both configurable using a key=value syntax (see [section 6](#) for the possible keys):

|                                                                                                                                                                                                                                      |                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <pre> \begin{prop}   \item[No Overlap,     align=flush,     display format=\textsc{#1},     ref format=\textit{#1}]     Nothing mental is     physical. \label{incomp} \end{prop} Is \ref{incomp} consistent with \ref{phys}? </pre> | <p>4 Nothing mental is physical.</p> <p>Is 4 consistent with <b>Physicalism</b>?</p> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|

Preset *prop styles* can be declared and used instead of configuring the `\items` one at a time:

|                                                                                                                                                                                         |                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <pre> \begin{prop}   \item[Nihilism,     style=thesis]     There is nothing.     \label{nihilism} \end{prop} Does \ref{nihilism} imply \ref{phys}, \ref{dual}, or both? Discuss. </pre> | <p>5 There is nothing.</p> <p>Does 5 imply <b>Physicalism</b>, (3), or both? Discuss.</p> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|

Shortcut commands—e.g., `\titem[⟨keys⟩]` for `\item[style=thesis, ⟨keys⟩]` (within `prop`)—can also be defined. The package loads with a range of predefined styles.

When the package is loaded with `\usepackage[equations]{propositions}`, first-level `\items` within `prop` use the same counter as equations. (This looks better with the `leqno` option to `\documentclass`.)

|                                                                                                                                                                                                 |                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <pre>\begin{equation}   \exists x (\text{Mental}(x)   \wedge \text{Physical}(x)) \end{equation} \begin{prop}   \item   There is overlap between   the mental and the physical. \end{prop}</pre> | $(6) \quad \exists x(\text{Mental}(x) \wedge \text{Physical}(x))$ <p>(7) There is overlap between the mental and the physical.</p> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|

The `\ptag` command (requires `amsmath`) is an analogue of `\item` (within `prop`) that works inside displayed math environments.

|                                                                                                                                               |                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <pre>\begin{equation}   \ptag{Monism} \label{mon}   \exists x \forall y (y = x) \end{equation} Is \ref{mon} compatible with \ref{dual}?</pre> | $\text{Monism} \quad \exists x \forall y (y = x)$ <p>Is <b>Monism</b> compatible with (3)?</p> |
|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|

## 5 The `prop` environment and `\pitem`

```
\begin{prop}[⟨keys⟩]
  ⟨environment content⟩
\end{prop}
```

Creates a displayed list of propositions. It is a standard L<sup>A</sup>T<sub>E</sub>X list, so by default its formatting will depend on the standard length parameters like `\itemsep` and `\topsep`, although these can be overridden by setting keys.

Within `prop`, `\pitem` (see below) creates labelled items. The ordinary `\item` command is still available for unlabelled items.

Any display math environment (`\equation`, `align`, etc.) used inside `prop` or `inlineprop` automatically increments the nesting level for its duration, so that `\ptag` inside such an environment behaves as if it were one level deeper.

The optional `⟨keys⟩` argument accepts the same keys as `\proppositions`<sup>P. 16</sup> (section 6), with effects local to this environment.

```
\begin{inlineprop}[⟨keys⟩]
  ⟨environment content⟩
\end{inlineprop}
```

Like `prop`, but does not create a list. Allows `\pitem` to be used outside list environments, e.g. for generating numbers at the beginning of paragraphs. Steps the `prop` counter and increments the nesting level. Accepts the same optional `⟨keys⟩` as `prop`.

Within `prop` and `inlineprop`, `\item` is redefined to act as the proposition-item command. Its optional argument is a comma-separated list of  $\langle key \rangle = \langle value \rangle$  pairs (see [section 6](#)). A bare string without `=` is treated as a proposition name.

When used without an optional argument (or without setting `name`, `counter`, or `style`), the style is determined by the `nameless style` key ([section 6](#)). By default this is `numbered`, which dispatches by nesting depth to `levelone`, `leveltwo`, ... via `\proplevelchoice`.

`\pitem[ $\langle keys \rangle$ ]` (deprecated)

Deprecated alias for `\item` within `prop` or `inlineprop`. Issues a warning on each use. Replace with `\item`.

`\ptag[ $\langle keys \rangle$ ]`

(Available only when `amsmath` is loaded.) Works inside displayed math environments like `equation` and `align`. Accepts the same keys as `\item` within `prop`, except that `align` has no effect (since positioning is controlled by the tag placement system).

## 6 Keys

Most keys are available in three contexts:

- In the optional argument of `\item` or `\ptag`: the key applies to that item only.
- In the optional argument of `prop` or `inlineprop`: list-dimension keys ([subsection 6.2](#)) take effect immediately for the whole list; other keys act as per-item defaults, applied before each item's own explicit keys.
- Via `\propoptions`<sup>[P.16](#)</sup> or as a `\usepackage` option: list-dimension keys take effect globally; item-level keys ([subsection 6.1](#)) act as persistent per-item defaults that are group-scoped (`\begingroup \propoptions{...} \endgroup` scopes the effect as usual).

When the same key is set in multiple contexts, the environment arg overrides `\propoptions`<sup>[P.16](#)</sup>; the style preset overrides both; and explicit item-level keys override everything. The `equations` key is global only ([subsection 6.5](#)).

### 6.1 Item-level keys

`name= $\langle text \rangle$`  (no default)

The proposition's name. A bare string (without `=`) in the key list is equivalent to `name= $\langle text \rangle$` .

`style= $\langle style \rangle$`  (no default)

A prop style, equivalent to a preset collection of keys. Styles can be declared using `\SetPropStyle`<sup>[P.9](#)</sup> or `\DeclareNumberedStyle`<sup>[P.10](#)</sup>, and several come predefined ([section 7](#)).

**counter**=*<name>* (no default)  
Counter to use. The counter is automatically stepped, and the item's **name** is set to `\the<name>` (though this can be overridden by **counter format** or **name**).

**counter format**=*<template>* (no default)  
How to display the counter value. Use **#1** for the counter name, e.g. **counter format**=`\roman{#1}`. When set, this is used instead of `\the<counter>` for generating the item's name.

**format**=*<template>* (no default)  
Formatting applied to the **name**: use **#1** for the argument, e.g. **format**=`\textbf{(#1)}`. Shorthand for setting both **display format** and **ref format**.

**display format**=*<template>* (no default)  
Format for displaying the name in the proposition's label. Does not affect cross-references.

**label format**=*<template>* (no default)  
Format applied to the *entire* assembled label, i.e. the name (after **display format**), shorthand, and gloss together. Use **#1** for the whole assembled text. For example, **label format**=**#1**: appends a colon after the complete label. Does not affect cross-references.

**ref format**=*<template>* (no default)  
Format for subsequent cross-references to this proposition. Does not affect the display.

**shorthand**=*<text>* (no default)  
An abbreviation displayed after the name. If present, the shorthand becomes the reference text: `\ref` produces the shorthand (formatted with **ref format**) rather than the full name.

**shorthand format**=*<template>* (initially `~[#1]`)  
Format for displaying the shorthand in the label.

**gloss**=*<text>* (no default)  
A parenthetical gloss displayed after the name. Does not affect cross-references.

**gloss format**=*<template>* (initially `~(#1)`)  
Format for displaying the gloss in the label.

**ref**=*<text>* (no default)  
Explicitly set the reference text, overriding what would be derived from **name**, **counter**, or **shorthand**.

**label**=*<label>* (no default)  
Equivalent to a trailing `\label{<label>}`.

`crefname`= $\langle type \rangle$  (no default)

When `cleveref` is loaded, assigns an arbitrary reference type to this proposition. For example, `crefname=lemma` causes `\cref` to use the names defined by `\crefname{lemma}{...}{...}` instead of the default `proposition` type. The  $\langle type \rangle$  must be known to `cleveref`; new types can be declared with `\crefname`.

`reset`= $\langle boolean \rangle$  (initially `true`)

When `true` (the default), processing this `\pitem` resets the sub-level counter one nesting depth below the current level (e.g. `numpropii` at level 1). Set `reset=false` to suppress this reset, so that sub-item numbering continues from where it left off across consecutive parent items. Can also be set in a style definition via `\SetPropStyle`<sup>→P.9</sup>.

`align`= $\langle value \rangle$  (no default)

How the label should be positioned. Possible values: `left` (standard left-aligned label, offset controlled by `\labelwidth` and `\labelsep`), `right` (right-aligned, like `enumerate`), `center` (centered within the label box), `flush` (aligned with left margin of item text), `nextline` (label on its own line), and `flush-nextline`. Has no effect inside `inlineprop` or `\ptag`. For left-margin alignment, use `labelindent=0em` instead.

## 6.2 List dimensions

These keys are effective in all three contexts. At item level, a dimension key applies only to the label of that specific item; at environment and global level it applies to the whole list.

`topsep`= $\langle length / length list \rangle$   
`partopsep`= $\langle length / length list \rangle$   
`itemsep`= $\langle length / length list \rangle$   
`parsep`= $\langle length / length list \rangle$   
`leftmargin`= $\langle length / length list \rangle$   
`rightmargin`= $\langle length / length list \rangle$   
`labelwidth`= $\langle length / length list \rangle$   
`labelsep`= $\langle length / length list \rangle$   
`itemindent`= $\langle length / length list \rangle$   
`listparindent`= $\langle length / length list \rangle$

Override the standard L<sup>A</sup>T<sub>E</sub>X list dimensions. Accept the same values as `\setlength`, including rubber lengths (e.g. `itemsep=4pt plus 2pt`). Each key may also take a comma-separated list of per-level values: e.g. `leftmargin={2.5em, 0em}`. The first value applies at level 1, the second at level 2, etc. Gaps in the list (e.g. `leftmargin={, 0em}`) cause the class default to be used for that level.

`labelindent`= $\langle length / length list \rangle$  (no default)

Positions the left edge of the label box at  $\langle length \rangle$  from the enclosing margin, adjusting `\labelwidth`, `\leftmargin`, `\labelsep`, or `\itemindent` as needed. (Since the value of any one of these dimensions can be derived from those of the other four, it never makes sense to set all five.)

**tightspacing** (no value)

Sets all vertical spacing to the compact defaults that the standard document classes use for level-three lists (**topsep** and **itemsep** to 2pt with stretch/shrink, **parsep** to 0pt, **partopsep** to 1pt). Individual dimension keys set afterward override.

**nosep** (no value)

Sets **\topsep**, **\itemsep**, and **\parsep** all to zero.

**continue**=*<boolean>* (initially **true**)

When **true**, a **prop** environment that immediately follows another **prop** (with no intervening paragraph text) replaces the normal **\topsep**-based inter-list space with **\itemsep** + **\parsep**, giving the visual appearance of a single continuous list.

**items**=*<n>* (initially 1)

Indicates that approximately *<n>* items are expected in this environment. Used together with **\propformlabel** to compute a preview label wide enough for the *<n>*th expected item, so that dimension expressions such as **leftmargin**=**\widthof{\propformlabel}**+0.5em reserve enough space for the widest label. Has no effect on actual item processing or counter stepping.

**\propformlabel**

Expands to the formatted label of the (approximately) **items**-th item in this environment, computed at **\begin{prop}** *before* the list dimensions are applied, so it can be used directly in dimension expressions. Updated after each **\item** to reflect the most recently output label. Typical use:

```
% \begin{prop}[items=20, leftmargin=\widthof{\propformlabel}+0.5em]
%
```

sizes the left margin to accommodate labels up to the 20th item.

## 6.3 Default styles

When **\pitem** or **\ptag** is used without an explicit **style**, one of the following defaults is selected based on whether a name/counter was given and whether **ptag** mode is active.

**named style**=*<style>* (initially **proposition**)

The style used when a **name** or **counter** is given but no explicit **style**.

**named ptag style**=*<style>* (initially empty)

If set, overrides **named style** for **\ptag** items only.

**nameless style**=*<style>* (initially **numbered**)

The style used when **\pitem** or **\ptag** has no optional argument (no name, counter, or style).

**nameless ptag style**=*<style>* (initially empty)

If set, overrides **nameless style** for **\ptag** items only.



## 6.4 Environment-level style and item defaults

Any key from [subsection 6.1](#) may appear in the optional argument of `prop` or `inlineprop`. It is then applied as a default before each `\item` that does not set that key explicitly. An item-level key always overrides the environment default.

**Important ordering note.** Style selection (which decides whether an item is “named” or “nameless”) is based solely on the item’s *own* optional argument, before environment defaults are applied. Consequently, `\begin{prop}[name=foo]\item bar` produces a *nameless*-style item with label (foo): the environment default `name=foo` is not seen during style resolution. To get the equivalent of `\item[name=foo]` (named style, bold), use `\begin{prop}[style=proposition, name=foo]\item bar`.

`style=<style>` (no default)

Sets a default style for all `\items` in this environment. The style’s list-dimension keys (`leftmargin`, `labelwidth`, etc.) take effect immediately for the whole list; item-level keys are applied as defaults for each `\item`. An explicit `style=` on an individual `\item` overrides this default. This key is also accepted by `\propoptions`, where it acts as a persistent default across all subsequent environments (group-scoped, overridden by the `env` arg, and overridden by per-item `style=`).

## 6.5 Equation numbering

`equations` (no value, **global only**)

Sets `nameless style=eqnum` and installs hooks so that the L<sup>A</sup>T<sub>E</sub>X and `amstex` displayed equation environments (such as `equation` and `align`) use the same formatting as the `equation` prop style. The `equation` style’s `display format` and `ref format` keys are used for generating equation numbers and cross-references to them. Redefining the `equation` style, e.g. with

```
| \SetPropStyle{equation}{format = [#1]}
```

will automatically update the hooks.

## 7 Prop styles

`\SetPropStyle{<name>}{<keys>}`

Defines or modifies a prop style for use with the `style` key. All `\pitem`<sup>P.5</sup> keys are accepted, plus the following:

`style=<parent>` (no default)

Inherit from a parent style. When an item is created, the parent’s settings are loaded first (recursively, if the parent itself has a parent), then this style’s own keys are applied on top.

The argument may be a macro which is expanded when the item is created: for example, a style with `style=\{<style1>\},\{<style2>\},\{<style3>\}` will resolve to `\{<style1>\}, \{<style2>\}, \{<style2>\}` depending on the nesting depth. The built-in `numbered` and `eqnum` styles use this mechanism.

`macro=<command>` (no default)

A new user macro, equivalent to `\pitem[style=<name>]`. Any further keys given to the macro are passed to `\pitem`.

If the style `<name>` already exists, `\SetPropStyle` modifies or adds keys. For example, `\SetPropStyle{proposition}{align=flush}` changes the alignment of the built-in `proposition` style while preserving its other settings.

|                                                                                                                                                                                                                                                                                                               |                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <pre> \SetPropStyle{angle}{   labelindent = 0em,   display format =     \textbf{\${\angle}\$#1\$\rangle\$},   ref format =     \${\angle}\$#1\$\rangle\$,   macro = \angitem } \begin{prop}   \angitem[Angle thesis]   Everything is angular. \end{prop} No further discussion of \lastref{} is needed.</pre> | <p><b>&lt;Angle thesis&gt;</b> Everything is angular.</p> <p>No further discussion of <b>&lt;Angle thesis&gt;</b> is needed.</p> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|

**`\DeclareNumberedStyle{<name>}[<keys>]`**

Creates a new L<sup>A</sup>T<sub>E</sub>X counter named `<name>` and a matching prop style with `counter=<name>`. All `\SetPropStyle`<sup>P.9</sup> keys are accepted, plus:

**`parent=<counter>`** (no default)

A parent counter; the new counter resets when the parent steps (same mechanism as `\numberwithin`). A dedicated `prop` counter (stepped by each `prop` and `inlineprop`) is available for non-persistent numbering.

|                                                                                                                                                                                              |                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <pre> \DeclareNumberedStyle{P} \begin{prop}   \item[counter=P] First premise.   \label{p1}   \item[counter=P] Second premise.   \label{p2} \end{prop} From \ref{p1} and \ref{p2}\ldots</pre> | <p><b>P1</b> First premise.</p> <p><b>P2</b> Second premise.</p> <p>From <b>P1</b> and <b>P2</b>...</p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|

**`\proplevelchoice{<item1, item2, ...>}`**

Picks the entry at the current nesting depth (1-indexed, clamped to the list length). At depth 0 (outside any `prop` environment), returns the first entry. This macro is fully expandable and can be used in the `name`, `format`, and `style` keys of `\SetPropStyle` to create depth-dependent styles.

## 7.1 Built-in styles

The following prop styles are predefined. Each entry shows the defining code (using user-facing commands) and a live example. All styles can be modified with `\SetPropStyle`<sup>P.9</sup>.

## Text styles

**plain** Unformatted text label.

```
| \SetPropStyle{plain}{format = #1}
```

**Claim** An unformatted item.

**proposition** Bold text label. Auto-selected when `\item` has a name but no style.

```
| \SetPropStyle{proposition}{format = \textbf{#1}}
```

**Thesis** A bold-formatted item.

**thesis** Small-caps name at the text margin (flush alignment). Shortcut: `\titem`.

```
| \SetPropStyle{thesis}{format = \textsc{#1},  
| align = flush, macro = \titem}
```

**HYPOTHESIS** A flush-aligned item.

**vignette** Italic name at the text margin with a colon in the label.

```
| \SetPropStyle{vignette}{ref format = \textit{#1},  
| align = flush, label format = \textit{#1:}}
```

***Example:*** A vignette item.

**bullet** Bullet symbol varying by depth (like `\itemize`). Shortcut: `\bitem`.

```
| \SetPropStyle{bullet}{align = center,  
| name = \proplevelchoice{\textbullet,  
| {\normalfont\textendash}, \textasteriskcentered,  
| \textperiodcentered},  
| display format = #1, macro = \bitem}
```

- A bullet item.

## Numbered styles with dedicated counters

**roman** Roman-numeral counter, resets per section. Shortcut: `\ritem`.

```
| \DeclareNumberedStyle{roman}[parent = section,  
| counter format = \roman{#1}, format = (#1), macro = \ritem]
```

- (i) A roman-numbered item.

**alph** Alphabetic counter, resets per section. Shortcut: `\aitem`.

```
| \DeclareNumberedStyle{alph}[parent = section,  
| counter format = \alph{#1}, format = (#1), macro = \aitem]
```

- (a) An alphabetic item.

## Level-specific styles

These styles are used by the `numbered` and `eqnum` dispatcher styles (below). They share the counters `numpropi`–`numpropv`, which reset automatically when an `\item` at the next lower level is processed.

`levelone` Level-1 numbered items: arabic in parentheses.

```
| \SetPropStyle{levelone}{counter = numpropi, format = (#1)}
```

`leveltwo` Level-2: lowercase alphabetic with dot.

```
| \SetPropStyle{leveltwo}{counter = numpropii,  
|   display format = #1., ref format = \parentref{#1}}
```

`levelthree` Level-3: lowercase roman in parentheses.

```
| \SetPropStyle{levelthree}{counter = numpropiii,  
|   display format = (#1), ref format = \parentref{.#1}}
```

`levelfour` Level-4: uppercase alphabetic with dot.

```
| \SetPropStyle{levelfour}{counter = numpropiv,  
|   display format = #1., ref format = \parentref{#1}}
```

`levelfive` Level-5: uppercase roman in parentheses.

```
| \SetPropStyle{levelfive}{counter = numpropv,  
|   display format = (#1), ref format = \parentref{.#1}}
```

`equation` Uses the `equation` counter. When the `equations` option is active, `\SetPropStyle{equation}{...}` automatically updates the `\tagform@` and `\ref` hooks for all equations.

```
| \SetPropStyle{equation}{counter = equation, format = (#1)}
```

## Dispatcher styles

These styles use `style` with `\proplevelchoice` to dispatch to a level-specific style at resolution time.

`numbered` The default nameless style. Dispatches to `levelone`–`levelfive` by nesting depth.

```
| \SetPropStyle{numbered}{style = \proplevelchoice{  
|   levelone, leveltwo, levelthree, levelfour, levelfive}}
```

`eqnum` Like `numbered`, but uses `equation` at level 1. Activated by the `equations` package option (which sets `nameless style = eqnum`).

```
\SetPropStyle{eqnum}{style = \proplevelchoice{
equation, leveltwo, levelthree, levelfour, levelfive}}
```

`hierarchical` Produces 1, 1.1, 1.1.1... numbering using `\parentref` and `counter` format. Defined via two helper styles:

```
\SetPropStyle{h-base}{counter = numpropi,
counter format = \arabic{#1}, format = #1}
\SetPropStyle{h-sub}{
counter = \proplevelchoice{numpropi, numpropii,
numpropiii, numpropiv, numpropv},
counter format = \arabic{#1},
format = \parentref{.#1}}
\SetPropStyle{hierarchical}{style = \proplevelchoice{
h-base, h-sub}}
```

## 8 Cross-referencing

Labels placed after `\item` items (within `prop`) work with the standard `\label/\ref` mechanism. The key difference from ordinary L<sup>A</sup>T<sub>E</sub>X references is that `\ref` produces *formatted* output: for example, `\textbf` might be applied to the name, or the number might be wrapped in parentheses. The formatting is controlled by the `format` key (or separately by `display format` and `ref format`).

```
\Ref{\label{}}
\Ref*{\label{}}
```

Titlecase variants of `\ref` and `\ref*`: uppercases the first letter of the formatted output. (For example, if `\ref{thesis}` produces ‘the Identity Theory’, then `\Ref{thesis}` produces ‘The Identity Theory’.) The starred form suppresses the hyperlink. Note: `\Ref` only affects references produced by this package (which are stored via `\propapply`); for other references, it behaves like `\ref`.

```
\nref{\label{}}
\nref*{\label{}}
\Nref{\label{}}
\Nref*{\label{}}
```

“Naked ref.” Outputs the bare reference content with all formatting stripped. If `\ref{premise}` produces ‘(P1)’, then `\nref{premise}` produces ‘P1’. The starred form suppresses the hyperlink. `\Nref` is the titlecase variant: it uppercases the first letter of the bare content.

`\nref` can be useful in the argument of `\item`, when the the name of one proposition should depend on that of another:

|                                                      |                                        |
|------------------------------------------------------|----------------------------------------|
| <code>\SetPropStyle{proposition}{format=(#1)}</code> | (Phys) Everything is physical.         |
| <code>\begin{prop}</code>                            |                                        |
| <code>\item[Phys]</code>                             | (Phys*) Almost everything is physical. |
| Everything is physical.                              |                                        |
| <code>\label{phys2}</code>                           |                                        |
| <code>\item[\nref{phys2}*]</code>                    | ((Phys*)) This one has two sets        |
| Almost everything is physical.                       | of parentheses, which is probably      |
| <code>\label{newphys2}</code>                        | not desired. (Phys*) avoided this      |
| <code>\item[\ref{phys2}*]</code>                     | by using <code>\nref</code> .          |
| This one has two sets of parentheses,                |                                        |
| which is probably not desired.                       |                                        |
| <code>\ref{newphys2}</code> avoided this by using    |                                        |
| <code> \nref </code> .                               |                                        |
| <code>\end{prop}</code>                              |                                        |

*Warning:* documents where the name of one item includes a reference to that of another, and there are further references to that item, will require multiple L<sup>A</sup>T<sub>E</sub>X runs to resolve all references. To save time, it is better to avoid long chains of dependencies of this sort.

```

\oref[⟨prefix⟩][⟨suffix⟩]{⟨label⟩}
\oref*[⟨prefix⟩][⟨suffix⟩]{⟨label⟩}
\Oref[⟨prefix⟩][⟨suffix⟩]{⟨label⟩}
\Oref*[⟨prefix⟩][⟨suffix⟩]{⟨label⟩}

```

“Ref with options.” Extends `\ref` by injecting a prefix and/or suffix *inside* the formatting. With one optional argument, `⟨suffix⟩` is appended; with two, `⟨prefix⟩` is prepended and `⟨suffix⟩` appended. For instance, if `\ref{premise}` produces ‘(P1)’, then `\oref[*]{premise}` produces ‘(P1\*)’ and `\oref[old~]{premise}` produces ‘(old~P1\*)’.

`\Oref` is the titlecase variant; the starred forms suppress the hyperlink.

`\oref` can also be useful in the name of `\items`, if one wants the display format for the modified item to depend on that originally used

|                                                      |                                    |
|------------------------------------------------------|------------------------------------|
| <code>\begin{prop}</code>                            |                                    |
| <code>\label{premise}</code>                         | (Phys*) This will use boldface and |
| <code>\item[style=plain,name=\oref[*]{phys2}]</code> | parentheses because the original   |
| This will use boldface and                           | referenced item did.               |
| parentheses because the original                     |                                    |
| referenced item did.                                 |                                    |
| <code>\end{prop}</code>                              |                                    |

Another handy use for `\oref` is in combination with `\nref` to refer to ranges:

|                                                    |                                    |
|----------------------------------------------------|------------------------------------|
| The first two numbered examples in                 | The first two numbered examples in |
| this document were                                 | this document were were (1–2).     |
| were <code>\oref[--\nref{atomism}]{atoms}</code> . |                                    |

```

\lastref[⟨prefix⟩]{⟨suffix⟩}
\Lastref[⟨prefix⟩]{⟨suffix⟩}

```

Produces a reference (with no hyperlink) to the most recently processed `\pitem` or `\ptag`, even without a `\label`. Useful for back-references in running text. With one argument, `⟨suffix⟩` is appended; with two, `⟨prefix⟩` is also prepended. Use `\lastref{}` for a plain reference. `\Lastref` is the titlecase variant.

`\nlastref`

`\nLastref`

Like `\lastref{}`, but returns the bare content without formatting. `\nLastref` is the titlecase variant.

`\parentref[⟨prefix⟩]{⟨suffix⟩}`

`\Parentref[⟨prefix⟩]{⟨suffix⟩}`

Inside a nested `prop` (or `inlineprop`), produces a formatted reference to the most recent item of the enclosing level. Same argument convention as `\lastref`<sup>→ P. 14</sup>. `\Parentref` is the titlecase variant.

`\nparentref`

`\nParentref`

Like `\parentref{}` but returns the bare content without formatting. Takes no arguments; simply output any desired suffix directly afterwards. `\nParentref` is the titlecase variant.

`\parentref` and `\nparentref` are useful for making subitems whose names derive from their parent's:

```
\DeclareNumberedStyle{inner}[
  counter format=\alph{inner},
  display format=#1.]
\begin{prop}
  \item[OI] Outer item.
  \label{outer2}
  \begin{prop}
    \item[style=inner,
      format=\parentref{.#1}]
    \label{dsub1}
    Ref: \ref{dsub1},
    naked: \nref{dsub1}.
    \item[style=inner,
      display format=#1.,
      ref format=\parentref{.#1}]
    \label{dsub2}
    Ref: \ref{dsub2},
    naked: \nref{dsub2}.
  \end{prop}
\end{prop}
```

**OI** Outer item.

**OI.a** Ref: **OI.a**, naked: a.

b. Ref: **OI.b**, naked: b.

The built-in `leveltwo` and `levelthree` styles have `ref format=\parentref{#1}` and `ref format=\parentref{.#1}`, respectively, so that if the parent references as `(P1)`, a `leveltwo` sub-item references as `(P1a)` and `\nref` returns just `a`.

## 8.1 How it works: `\propapply`

`\propapply{⟨template⟩}{⟨content⟩}`

Internally, each reference is stored in the `.aux` file as `\propapply{⟨template⟩}{⟨content⟩}`. The `⟨template⟩` contains formatting with the placeholder `\propfmtarg`<sup>→ P. 16</sup> where content appears. At reference time, `\propapply` evaluates the template with `\propfmtarg` bound to `⟨content⟩`. The `\oref`<sup>→ P. 14</sup> and `\nref`<sup>→ P. 13</sup> commands work by locally redefining `\propapply`.

In normal use, you need not interact with `\propapply` directly.

#### `\propfmtarg`

Placeholder used inside templates; expands to the content argument of the enclosing `\propapply`<sup>→ P. 15</sup>.

## 9 Package options

#### `\proptions{<keys>}`

Sets package-level keys (see [section 6](#)). List-dimension keys take effect globally. Item-level keys ([subsection 6.1](#))—including `style=`—act as persistent per-item defaults: they are applied before each `\item` in every subsequent `prop` or `inlineprop`, but can be overridden by the environment arg or by explicit item-level keys. In particular, `\proptions{style=thesis}` makes every subsequent `prop` env use `thesis` style by default. Because the underlying storage uses a local L<sup>A</sup>T<sub>E</sub>X3 variable, the effect is group-scoped: `\begingroup \proptions{style=thesis} ... \endgroup` applies the default only within the group. The same keys can also be set in the optional argument of `prop` or `inlineprop` (local to that environment), or as options to `\usepackage` (global, but group-scoped from the preamble group).

## 10 Compatibility

The `propositions` package is designed to work with `hyperref`, `cleveref`, and `amsmath`. `amsmath` is required for `\ptag` and the `equations` option. With `cleveref` loaded, all proposition items are assigned to a default proposition “reference type”; the `crefname` key can override this.

Recommended load order:

```
\usepackage{hyperref}  
\usepackage{amsmath} % if using \ptag  
\usepackage{propositions}  
\usepackage{cleveref} % if used
```

## 11 Known issues

When using `\ptag` with a named counter (e.g. `\ptag[counter=P]`) inside an `amsmath` equation environment, `hyperref` may emit warnings of the form:

```
pdfTeX warning: destination with the same  
identifier (name{equation.N}) has been already  
used, duplicate ignored
```

These warnings are harmless and do not affect the correctness of cross-references.